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clc; clearvars;

f = 433e6; % frequency in Hertz
Re = 6371e3; % Earth radius in metres
h = 450e3; % satellite altitude above Earth's surface in metres
elevation_angles = 5:5:90; % elevation angle range in degrees

P_t = 20; % includes HPA, Power of the transmitter in dBm
G_t = 1; % gain of the antenna, in dBi
L_txsystem = 1; %losses due to feeding lines, connectors, filters, HPA etc in dB
L_fspl = 0; %free space path loss (calculation below) in dB
L_atm = 0; %atmospheric, rain, clouds, fog, ionosphere attenuation + (Diffusion-
due to obstacles- neglect) + (Multipath fading- only for terrestrial- neglect) in dB
L_plrz = 3; %polarization mismatch, effectively 0 for circularly polarised antennas
in dB
L_point = 1.5; % pointing losses, add both reciever and transmitter pointing losses
in dB

G_r = 9; % gain of ground station in dBi
G_lna_ext = 0; %gain of external lna, see below for effect on F in dBi
L_rxsystem = 1.5; %losses due to feeding lines, connectors, filters, LNA,
demodulators etc in dB

S_datasheet = -132; %sensitivity on datasheet (already includes the internal LNA
gain effect an noise spectrum density change) in dBm
T_cold = 200; %At zenith
T_hot = 1000; %at horizon
T_ant = 0; %Thermal temperature, Tsky(atmospheric thermal temp) + Tgnd(due to sie
and back lobes) in K, Calculation below
B_datasheet = 125e3; %Bandwidth of reciever in Hz
SNRrequired_datasheet = -17.5; %required signal to noise ratio for the reciever to
be able to demod the signal in dB

U_bitrate = 980e-6; %bitrate that WE pass, megabits per second
EbNo_threshold = 7.1; %minimum Eb/No required by the reciever, in dB

% Preallocate arrays
LINKMARGINP_via_sensitivity = zeros(size(elevation_angles));
LINKMARGINP_via_ebno = zeros(size(elevation_angles));
r_all = zeros(size(elevation_angles));
L_fspl_all = zeros(size(elevation_angles));
Ts_actual_all = zeros(size(elevation_angles));
CNo_all = zeros(size(elevation_angles));
EbNo_all = zeros(size(elevation_angles));

for i = 1:length(elevation_angles)
    el = elevation_angles(i); % current elevation angle
    fprintf('\n=== Elevation Angle: %.1f° ===\n', el);

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    r = sqrt((Re + h)^2 - (Re * cosd(el))^2) - Re * sind(el); % Compute slant range
(distance from ground station to satellite)
    EIRP = P_t + G_t - L_txsystem; %effective output power in dBm
    L_fspl = 20*log10(4*pi*r*f/299792458);
    L_prop = L_fspl + L_atm; %propagation losses

    Pr = EIRP - L_prop - L_plrz - L_point + G_r + G_lna_ext - L_rxsystem;
    C = Pr; %recieved power to digital system at reciever end, in dBm
    T_ant = T_cold + (T_hot - T_cold) * (1 - sind(el)); %using Thot an Tcold to
model elevation differences
    N = S_datasheet - SNRrequired_datasheet; %noise power
    NF = N + 174 - 10*log10(B_datasheet); %noise figure at datasheet
    F = 10^(NF/10); %noise factor at datasheet

    % if external lna connected - F = Flna + (Frx - 1)/G_lna_ext;

    Tr = (F-1)*290; %Thermal temp at reciever end
    Ts_actual = T_ant + Tr; %Thermal temp of system including Tatm and Tr (entire
system)
    Ts_spec = F*290; %Thermal temp use by mnufacturer to calculate
correction = 10*log10(Ts_actual/Ts_spec); %change due to Tatm

    S_actual = S_datasheet + correction; %actual sensitivity of reciever system
    Psensitivity = S_actual;
    LINKMARGINP_via_sensitivity(i) = Pr - Psensitivity;

    No = 1.380649e-23 * Ts_actual; %Noise spectral density in W/Hz
    No = 10*log10(No); %nsd in dBW/Hz

    C_ = C - 30; %C in dBW
    CNo = C_ - No; %received carrier power/noise spectral density in dB-Hz
    EbNo_calculated = CNo - 10*log10(U_bitrate) - 60; %received EbNo in dB, 60
subtract to convert from Mbps to bps
    LINKMARGINP_via_ebno(i) = EbNo_calculated - EbNo_threshold;

    % Store results for table
    r_all(i) = r;
    L_fspl_all(i) = L_fspl;
    Ts_actual_all(i) = Ts_actual;
    CNo_all(i) = CNo;
    EbNo_all(i) = EbNo_calculated;

    % Print key values for this elevation
    fprintf('Slant range (r): %.2f km\n', r/1e3);
    fprintf('Free-space path loss (L_fspl): %.2f dB\n', L_fspl);
    fprintf('System temperature (Ts_actual): %.2f K\n', Ts_actual);
    fprintf('C/N0: %.2f dB-Hz\n', CNo);
    fprintf('Eb/No: %.2f dB\n', EbNo_calculated);
    fprintf('Link Margin (via Eb/No): %.2f dB\n', LINKMARGINP_via_ebno(i));
    fprintf('-----\n');

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end

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=== Elevation Angle: 5.0° ===  
Slant range (r): 1943.68 km  
Free-space path loss (L_fsp1): 150.95 dB  
System temperature (Ts_actual): 2707.98 K  
C/N0: 36.32 dB-Hz  
Eb/No: 6.41 dB  
Link Margin (via Eb/No): -0.69 dB  
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=== Elevation Angle: 10.0° ===  
Slant range (r): 1569.57 km  
Free-space path loss (L_fsp1): 149.09 dB  
System temperature (Ts_actual): 2638.78 K  
C/N0: 38.29 dB-Hz  
Eb/No: 8.38 dB  
Link Margin (via Eb/No): 1.28 dB  
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=== Elevation Angle: 15.0° ===  
Slant range (r): 1293.07 km  
Free-space path loss (L_fsp1): 147.41 dB  
System temperature (Ts_actual): 2570.65 K  
C/N0: 40.09 dB-Hz  
Eb/No: 10.18 dB  
Link Margin (via Eb/No): 3.08 dB  
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=== Elevation Angle: 20.0° ===  
Slant range (r): 1089.70 km  
Free-space path loss (L_fsp1): 145.92 dB  
System temperature (Ts_actual): 2504.09 K  
C/N0: 41.69 dB-Hz  
Eb/No: 11.78 dB  
Link Margin (via Eb/No): 4.68 dB  
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=== Elevation Angle: 25.0° ===  
Slant range (r): 938.75 km  
Free-space path loss (L_fsp1): 144.63 dB  
System temperature (Ts_actual): 2439.61 K  
C/N0: 43.10 dB-Hz  
Eb/No: 13.19 dB  
Link Margin (via Eb/No): 6.09 dB  
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=== Elevation Angle: 30.0° ===  
Slant range (r): 824.96 km  
Free-space path loss (L_fsp1): 143.51 dB  
System temperature (Ts_actual): 2377.70 K  
C/N0: 44.33 dB-Hz  
Eb/No: 14.42 dB  
Link Margin (via Eb/No): 7.32 dB  
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=== Elevation Angle: 35.0° ===  
Slant range (r): 737.78 km  
Free-space path loss (L_fsp1): 142.54 dB  
System temperature (Ts_actual): 2318.84 K  
C/N0: 45.41 dB-Hz  
Eb/No: 15.50 dB  
Link Margin (via Eb/No): 8.40 dB  
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=== Elevation Angle: 40.0° ===  
Slant range (r): 669.99 km  
Free-space path loss (L_fsp1): 141.70 dB  
System temperature (Ts_actual): 2263.47 K  
C/N0: 46.35 dB-Hz  
Eb/No: 16.44 dB
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Link Margin (via Eb/No): 9.34 dB
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=== Elevation Angle: 45.0° ===
Slant range (r): 616.67 km
Free-space path loss (L_fsp1): 140.98 dB
System temperature (Ts_actual): 2212.02 K
C/N0: 47.17 dB-Hz
Eb/No: 17.26 dB
Link Margin (via Eb/No): 10.16 dB
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=== Elevation Angle: 50.0° ===
Slant range (r): 574.38 km
Free-space path loss (L_fsp1): 140.36 dB
System temperature (Ts_actual): 2164.87 K
C/N0: 47.88 dB-Hz
Eb/No: 17.97 dB
Link Margin (via Eb/No): 10.87 dB
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=== Elevation Angle: 55.0° ===
Slant range (r): 540.74 km
Free-space path loss (L_fsp1): 139.84 dB
System temperature (Ts_actual): 2122.38 K
C/N0: 48.49 dB-Hz
Eb/No: 18.58 dB
Link Margin (via Eb/No): 11.48 dB
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=== Elevation Angle: 60.0° ===
Slant range (r): 514.02 km
Free-space path loss (L_fsp1): 139.40 dB
System temperature (Ts_actual): 2084.88 K
C/N0: 49.01 dB-Hz
Eb/No: 19.10 dB
Link Margin (via Eb/No): 12.00 dB
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=== Elevation Angle: 65.0° ===
Slant range (r): 493.01 km
Free-space path loss (L_fsp1): 139.03 dB
System temperature (Ts_actual): 2052.66 K
C/N0: 49.44 dB-Hz
Eb/No: 19.53 dB
Link Margin (via Eb/No): 12.43 dB
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=== Elevation Angle: 70.0° ===
Slant range (r): 476.81 km
Free-space path loss (L_fsp1): 138.74 dB
System temperature (Ts_actual): 2025.95 K
C/N0: 49.79 dB-Hz
Eb/No: 19.88 dB
Link Margin (via Eb/No): 12.78 dB
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=== Elevation Angle: 75.0° ===
Slant range (r): 464.78 km
Free-space path loss (L_fsp1): 138.52 dB
System temperature (Ts_actual): 2004.96 K
C/N0: 50.06 dB-Hz
Eb/No: 20.14 dB
Link Margin (via Eb/No): 13.04 dB
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=== Elevation Angle: 80.0° ===
Slant range (r): 456.47 km
Free-space path loss (L_fsp1): 138.37 dB
System temperature (Ts_actual): 1989.86 K
C/N0: 50.25 dB-Hz
Eb/No: 20.33 dB

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Link Margin (via Eb/No): 13.23 dB

==== Elevation Angle: 85.0° ====

Slant range (r): 451.60 km

Free-space path loss (L_fspl): 138.27 dB

System temperature (Ts_actual): 1980.75 K

C/N0: 50.36 dB-Hz

Eb/No: 20.45 dB

Link Margin (via Eb/No): 13.35 dB

==== Elevation Angle: 90.0° ====

Slant range (r): 450.00 km

Free-space path loss (L_fspl): 138.24 dB

System temperature (Ts_actual): 1977.70 K

C/N0: 50.40 dB-Hz

Eb/No: 20.48 dB

Link Margin (via Eb/No): 13.38 dB

% Produce table of results

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Results = table(elevation_angles.', r_all.'/1e3, L_fspl_all.', Ts_actual_all.', ...
    CNo_all.', EbNo_all.', LINKMARGINP_via_sensitivity.', LINKMARGINP_via_ebno.',
    ...
    'VariableNames', {'Elevation_deg', 'SlantRange_km', 'L_fspl_dB', 'Ts_K', ...
    'CNo_dBHz', 'EbNo_dB', 'LinkMargin_Sens_dB', 'LinkMargin_EbNo_dB'});
```

disp(Results);

Elevation_deg	SlantRange_km	L_fspl_dB	Ts_K	CNo_dBHz	EbNo_dB	LinkMargin_Sens_dB	LinkMargin_EbNo_dB
5	1943.7	150.95	2708	36.323	6.4104	2.8784	-0.68
10	1569.6	149.09	2638.8	38.292	8.3797	4.8477	1.2
15	1293.1	147.41	2570.6	40.089	10.177	6.6445	3.0
20	1089.7	145.92	2504.1	41.689	11.777	8.2447	4.0
25	938.75	144.63	2439.6	43.097	13.185	9.6532	6.0
30	824.96	143.51	2377.7	44.331	14.419	10.887	7.3
35	737.78	142.54	2318.8	45.41	15.498	11.966	8.3
40	669.99	141.7	2263.5	46.352	16.44	12.908	9.3
45	616.67	140.98	2212	47.173	17.26	13.728	10
50	574.38	140.36	2164.9	47.883	17.971	14.439	10
55	540.74	139.84	2122.4	48.494	18.581	15.049	11
60	514.02	139.4	2084.9	49.011	19.099	15.567	11
65	493.01	139.03	2052.7	49.441	19.529	15.997	12
70	476.81	138.74	2025.9	49.789	19.876	16.344	12
75	464.78	138.52	2005	50.056	20.143	16.611	13
80	456.47	138.37	1989.9	50.245	20.333	16.801	13
85	451.6	138.27	1980.7	50.358	20.446	16.914	13
90	450	138.24	1977.7	50.396	20.484	16.951	13

% Plot link margins vs elevation angle

figure;

plot(elevation_angles, LINKMARGINP_via_sensitivity, '-o', 'LineWidth', 1.5); hold on;

plot(elevation_angles, LINKMARGINP_via_ebno, '-s', 'LineWidth', 1.5);

grid on;

xlabel('Elevation Angle (°)');

```
ylabel('Link Margin (dB)');  
title('Link Margin vs Elevation Angle');  
legend('Via Sensitivity', 'Via Eb/No', 'Location', 'best');
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